Optimization

CS5491: Artificial Intelligence
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Content Credits: Prof. Wei's CS4486 Course and Prof. Boddeti's AI Course

ANNOUNCEMENTS

Midterm: Thursday, 22 October 2026

20% of the course grade; attend your enrolled session.

Printed-materials-only open book; no electronic devices.

Project proposal: 15 October, 11:59 pm BJT.

Use the practice paper and solutions to prepare.

TODAY: MODELING OPTIMIZATION

LP and IP are examples of a more general framework.

Identify the decision variables, feasible set, and objective.

Distinguish a feasible point, an optimal solution, and the optimal value.

Formulate examples and check what a solver returns.

AN OPTIMIZATION MODEL

$$\min_{\mathbf{x} \in \mathcal{F}} f(\mathbf{x})$$

Component	Meaning
Decision variables $\mathbf{x}$	Quantities we choose.
Feasible set $\mathcal{F}$	Choices satisfying all domain restrictions and constraints.
Objective $f : \mathcal{F} \to \mathbb{R}$	The value we want to minimize.

LP example: $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$, $\mathcal{F} = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$.

Given: $A, \mathbf{b}, \mathbf{c}$. Choose: $\mathbf{x}$.

VARIABLE DOMAINS AND CONSTRAINTS

Variable type	Example domain	Example decision
Continuous	$a \in \mathbb{R}$	A regression slope.
Discrete	$z \in \{0, 1\}$	Whether to select an item.
Mixed	$(a, z) \in \mathbb{R} \times \{0, 1\}$	A quantity and an on/off choice.

Additional constraints can further restrict these choices.

Unconstrained optimization here means $\mathcal{F} = \mathbb{R}^n$.

Example: requiring a nonnegative slope gives $\mathcal{F} = [0, \infty)$.

OPTIMAL SOLUTIONS AND VALUES

When a minimum is attained:

$$\mathbf{x}^* \in \operatorname{argmin}_{\mathbf{x}\in\mathcal{F}} f(\mathbf{x}), \qquad f^* = \min_{\mathbf{x}\in\mathcal{F}} f(\mathbf{x}) = f(\mathbf{x}^*).$$

argmin: the set of all optimal choices.

f^*: the optimal objective value, a number.

Example: minimize t^2 over $\mathcal{F} = \{-1, 1\}$.

$$\operatorname{argmin}_{t\in\mathcal{F}} t^2 = \{-1, 1\}, \qquad f^* = 1.$$

DOES AN OPTIMAL SOLUTION EXIST?

Problem	Outcome
$\min t, t \in [0, 1]$	Attained: $t^* = 0$, value 0.
$\min t, t \in \mathbb{R}$	Unbounded below.
$\min t, t \geq 1$ and $t \leq 0$	Infeasible: no allowed choice.

MINIMIZATION AND MAXIMIZATION

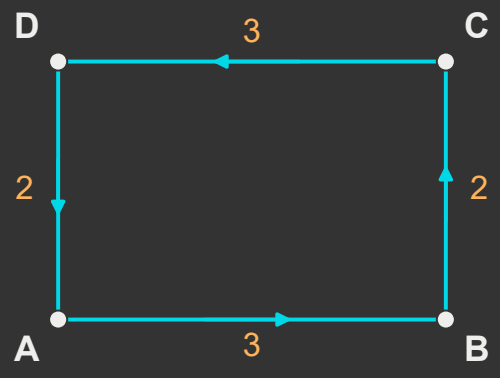
Define $g(\mathbf{x}) = -f(\mathbf{x})$ on the same feasible set.

$$\operatorname{argmin}_{\mathbf{x}\in\mathcal{F}} f(\mathbf{x}) = \operatorname{argmax}_{\mathbf{x}\in\mathcal{F}} g(\mathbf{x}).$$

The optimal choices are the same. When attained, the optimal values satisfy $g^* = -f^*$.
Example: $\mathcal{F} = \{2, 5\}$, $f(t) = t$, and $g(t) = -t$.

$$t^* = 2 \text{ in both problems;} \qquad f^* = 2, \quad g^* = -2.$$

TRAVELING SALESMAN PROBLEM



Data: distances $d(i, j)$.

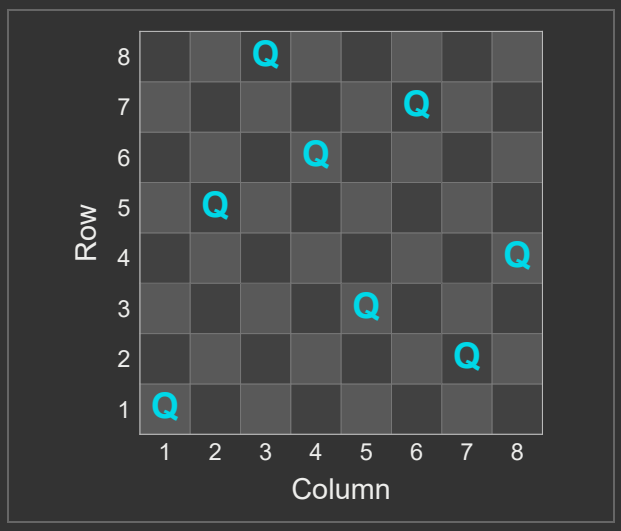
Decision: a visiting order $\mathbf{x} = (x_1, \ldots, x_n)$.

Feasible set: all permutations of the n cities.

$$f(\mathbf{x}) = \sum_{k=1}^{n-1} d(x_k, x_{k+1}) + d(x_n, x_1).$$

The closing edge returns to the start; this tour has length 10.

8 QUEENS AS A FEASIBILITY PROBLEM



x_i: row of the queen in column i.

$$x_i \in \{1, \ldots, 8\}.$$

For every $1 \leq i < j \leq 8$:

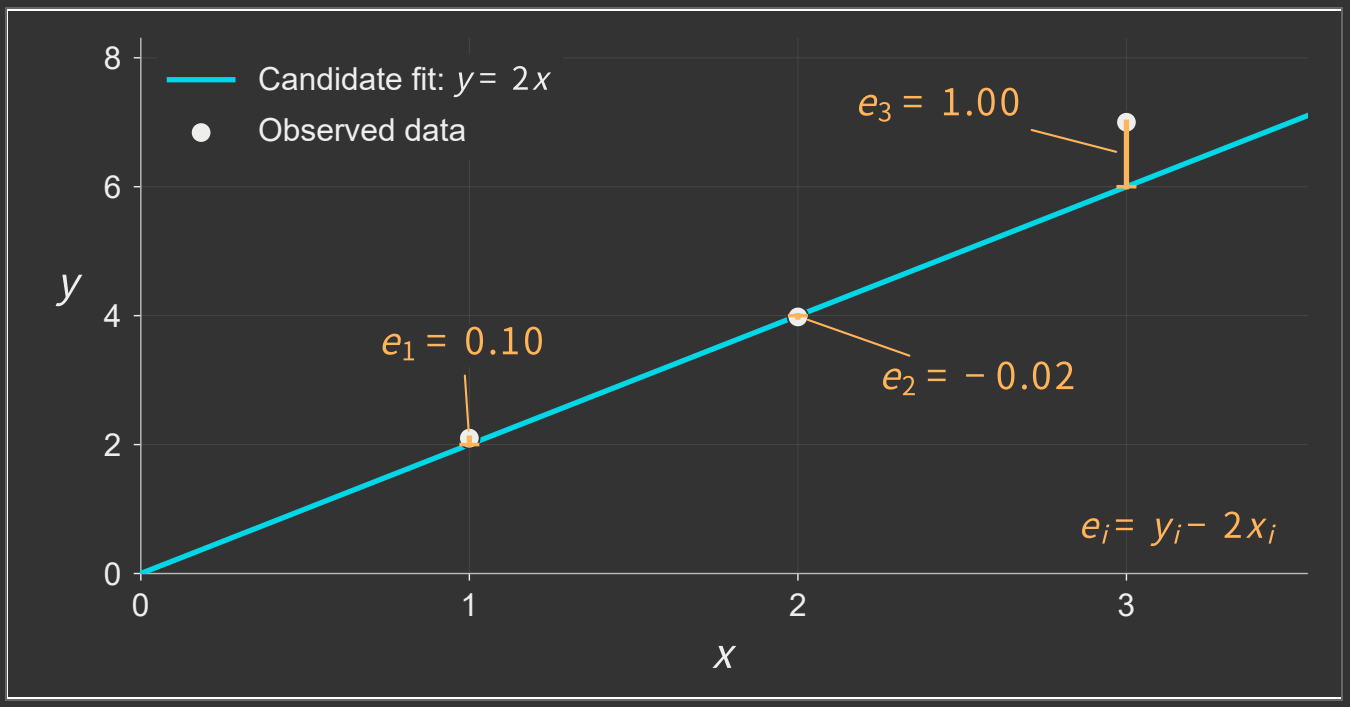
$$x_i \neq x_j,$$

$$|x_i - x_j| \neq |i - j|.$$

One queen per column is built into the representation. Minimize the constant objective $f(\mathbf{x}) = 0$ over these constraints. Every feasible placement is optimal.

FITTING A LINE THROUGH THE ORIGIN

Given observations: $(1, 2.1), (2, 3.98), (3, 7.0)$.



Decision: slope $a \in \mathbb{R}$. Prediction: $\hat{y}_i = ax_i$.
Residual: $e_i(a) = y_i - ax_i$. The pictured $a = 2$ is a candidate.

CHOOSE AN ERROR OBJECTIVE

What objective would make the residuals $e_i(a) = y_i - ax_i$ small?

Absolute error

$$L_1(a) = \sum_{i=1}^{3} |y_i - ax_i|.$$

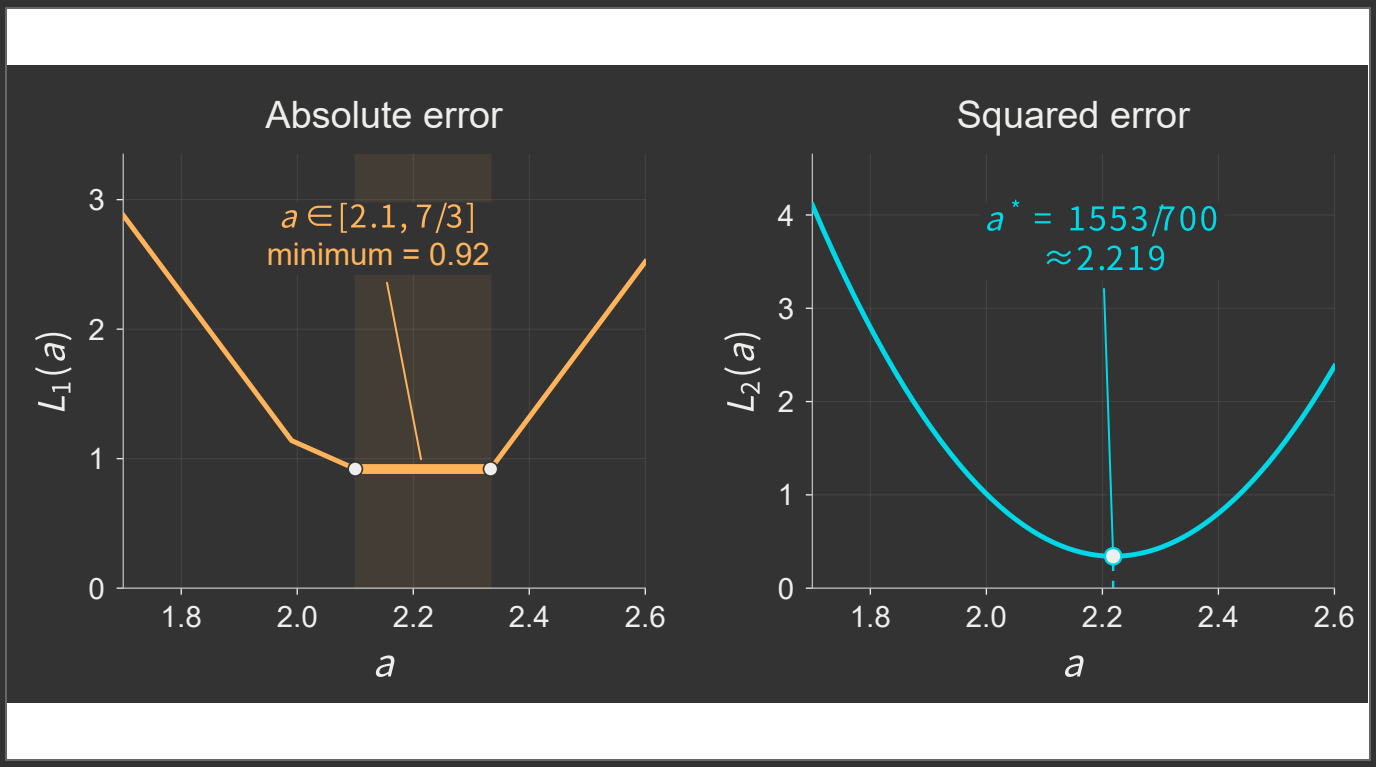
Squared error

$$L_2(a) = \sum_{i=1}^{3} (y_i - ax_i)^2.$$

At $a = 2$, the residuals are $(0.10, -0.02, 1.00)$: $L_1(2) = 1.12$ and $L_2(2) = 1.0104$.

Compare candidates using the same loss; these two values have different scales.

THE LOSS CHANGES THE BEST FIT



Absolute error has an interval of best slopes.
Squared error has one best slope.
Both are convex; convexity does not guarantee uniqueness.

RECOGNIZING PROBLEM STRUCTURE

Model	Structure that matters
Linear program	Linear objective and linear constraints; continuous variables.
Integer / mixed-integer program	Some or all variables must take integer values.
Convex minimization	Convex objective over a convex feasible set.

LP is a special case of convex optimization.

Integer restrictions generally make a feasible set nonconvex.

Both regression losses are nonlinear but convex.

SOLVING AND CHECKING A MODEL

1. Formulate the variables, constraints, and objective.
2. Choose a method suited to the model's structure.
3. Solve and inspect the reported status.
4. Check feasibility and recompute the objective.
5. Interpret the result in the original problem.

A feasible point need not be optimal. A solver may also report infeasibility, unboundedness, or an unfinished search.

CHECK THE MODEL

Fit $\hat{y}_i = ax_i$ to the same observations, requiring $a \geq 0$ and minimizing squared error.

Question	Answer
What is chosen?	The slope a.
What data are given?	The observations (x_i, y_i).
What is the feasible set?	$\mathcal{F} = [0, \infty)$.
What is the objective?	$f(a) = \sum_{i=1}^{3} (y_i - ax_i)^2$.
Is this a linear program?	No: the objective is quadratic in a.

KEY POINTS

Separate what is chosen from what is known.

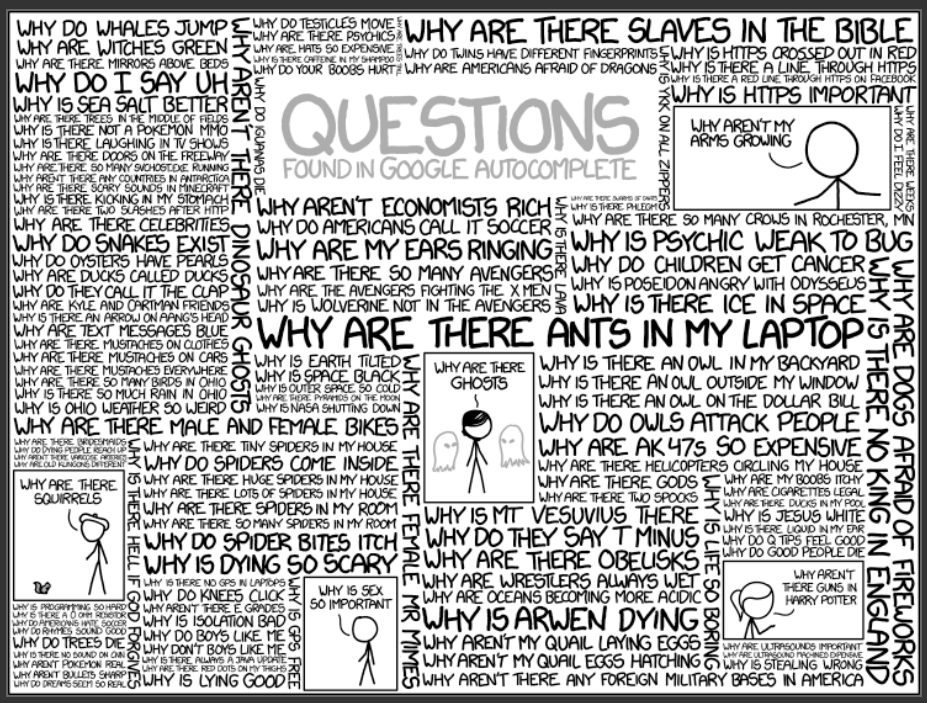
Feasibility, optimality, and the optimal value are different concepts.

The representation and objective determine the optimization problem.

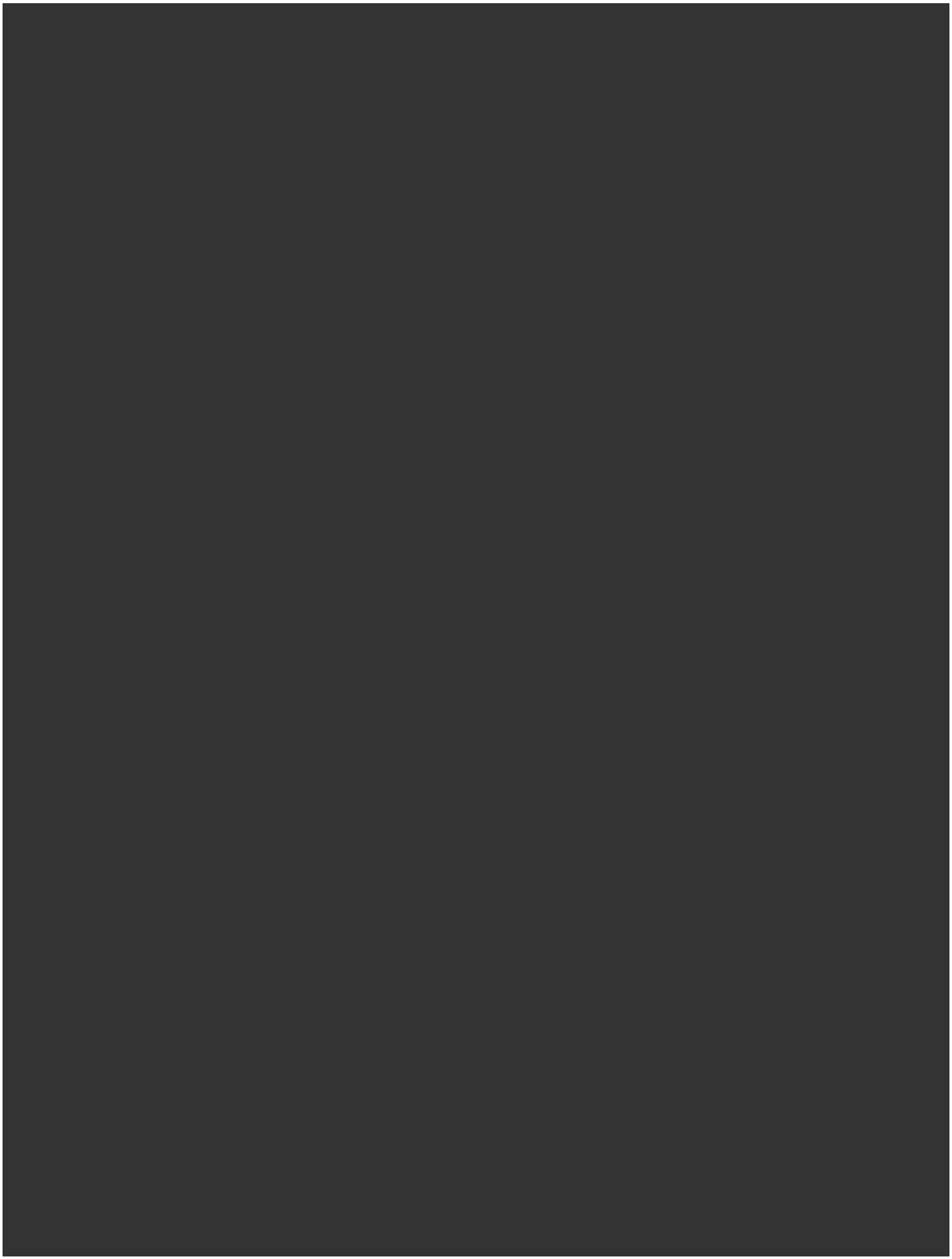
Check a solver's result before interpreting it.

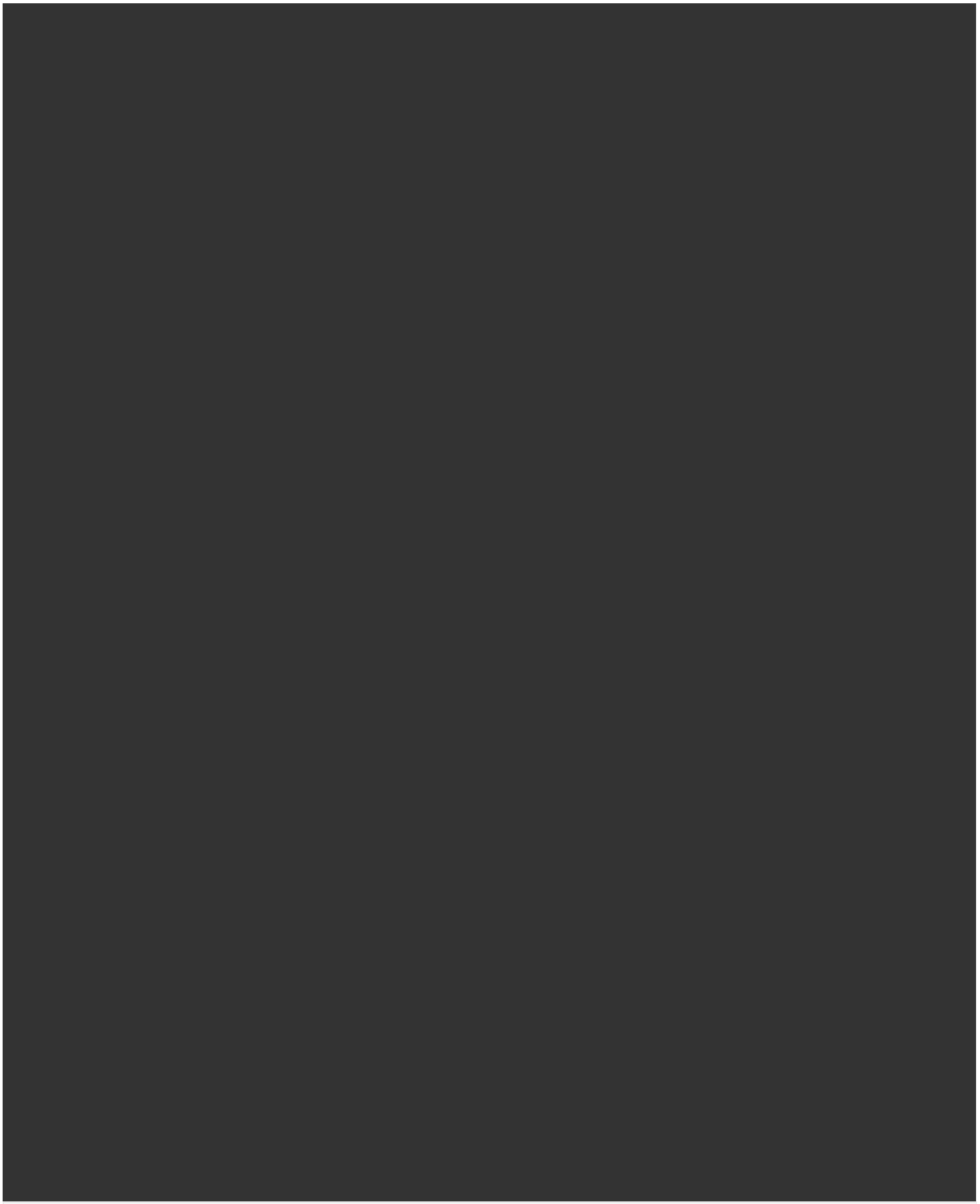
Next: how do convex sets and convex objectives help us solve these models?

Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE THERE GHOSTS
WHY ARE THERE SQUIRRELS
WHY AREN'T MY ARMS GROWING
WHY ARE THERE GUNS IN HARRY POTTER
WHY IS SEX SO IMPORTANT
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SHOWSTOPPER RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SUSHIES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARSITY PRESENT
WHY ARE OLD KUNGONS DIFFERENT
WHY IS THERE HELL IF GOD FORGIVES
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO ISLANDERS DIE
WHY AREN'T THERE DINOSAUR GHOSTS
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY TABS
WHY IS LYING GOOD
WHY IS THERE LAVA
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS LIFE SO BORING
WHY IS THERE ICE IN SPACE
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS THERE ENLIGHT
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS CROSSED OUT IN RED
WHY IS HTTPS IMPORTANT
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY ARE THERE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND PHONES DEFAME
WHY IS STEALING WRONG

XKCD





Speaker notes

